Machine Learning for Decision Making and Prediction

Machine Learning is a branch of AI where a computer learns patterns from data and then makes decisions or predictions without being explicitly programmed for every rule.

How it works

- Collect data
- Train a model on that data
- The model learns patterns
- It predicts outcomes for new data

In Machine Learning and Statistics, variables are mainly divided into Independent Variables and Dependent Variables. These define how a model learns relationships in data.

Independent Variable (X):

An independent variable is the input feature that influences or explains another variable.

It is called independent because its value does not depend on another variable in the model.

Dependent Variable (Y):

A dependent variable is the output or result that depends on the independent variables.

It is the variable we want to predict or explain.

Supervised Learning:

Supervised learning is a type of machine learning where the model is trained using labeled data, meaning the correct output (target variable) is already provided.

The model learns the relationship between input variables and the target variable and then uses that pattern to predict outputs for new data.

Training a model using labeled data (inputs + correct outputs) so it can predict future results.

1. Regression

Used when the output is continuous numerical values.

- House price prediction
- Sales forecasting
- Temperature prediction

2. Classification

Used when the output is categories or classes.

- Spam / Not spam
- Fraud / Not fraud

- Churn / Not churn

Supervised Algorithms:

- An algorithm is a set of rules or instructions designed to solve a specific problem.
- It processes input data to produce output or insights.

Linear Regression:

Linear Regression is used when the dependent variable (target) is continuous numeric data.

It finds a linear relationship between independent variables (X) and dependent variable (Y).

The model learns a line that best fits the data.

1) Data Input

2) Exploratory Data Analysis (EDA)-EDA is used to understand the dataset.

3) Data Cleaning- Before transformation we usually do data cleaning.

4) Data Transformation- Data transformation converts raw data into a form suitable for modeling.

5) Feature Selection- Before choosing algorithm we often choose important variables.(correlation matrix)

6) Selection of ML Algorithm- Since the target is continuous numeric, we choose Linear Regression.

7) Data Standardization / Normalization-

8) Train-Test Split- We divide dataset into two parts.(Train model on training data)

9) Model Training-(Train the linear regression model.)

10) Model Evaluation- Use the trained model to predict values.

11) Model Accuracy

For linear regression we evaluate using metrics.

R^2 Score

Measures how well model explains variance.

11) Model Deployment-(After evaluation, we deploy the model so applications can use it.)

12) Logging and Monitoring-(After deployment we monitor the system)

Logging tracks:

- prediction errors
- system failures
- model drift

2. Logistic Regression

- Logistic Regression is used when the target variable is categorical (classification problem).
- Instead of predicting numbers, it predicts probabilities between 0 and 1.
- These probabilities are converted into classes.

Use Cases

- Spam detection
- Fraud detection
- Disease prediction
- Customer churn prediction

Residual in Linear Regression

Residual is the difference between the actual value and the predicted value from the regression line.

Regression Analysis:

Regression Analysis is a statistical and machine learning technique used to study the relationship between a dependent variable and one or more independent variables in order to predict outcomes or understand influence.

Linear Regression Equation: $y = mx + c$
y = dependent variable (target we predict)
X = independent variable
m = slope (how much y changes when x increases)
c → intercept (value of y when x = 0)

TSS (Total Sum of Squares)
TSS measures total variation in the actual data.
It represents how spread the data is from the mean.

RSS (Residual Sum of Squares)
RSS measures error in predictions.

R^2 Statistic (Coefficient of Determination):
How much variation in the dependent variable is explained by the model.

$$R^2 = 1 - \frac{RSS}{TSS}$$

$$R^2 = \frac{RSS}{TSS}$$

R^2 **always increases when more variables are added**, even if they are useless.

Adjusted R^2--Corrected R^2 considering number of variables.
Adjusted R^2 adjusts this value by penalizing unnecessary predictors to prevent overfitting.